Advanced Computer-Aided VLSI System Design

Network on Chip (NoC)

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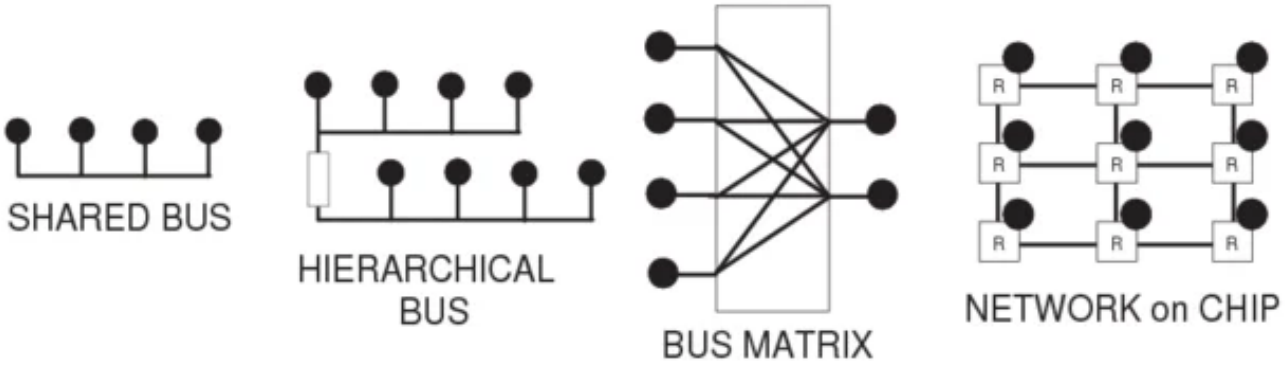


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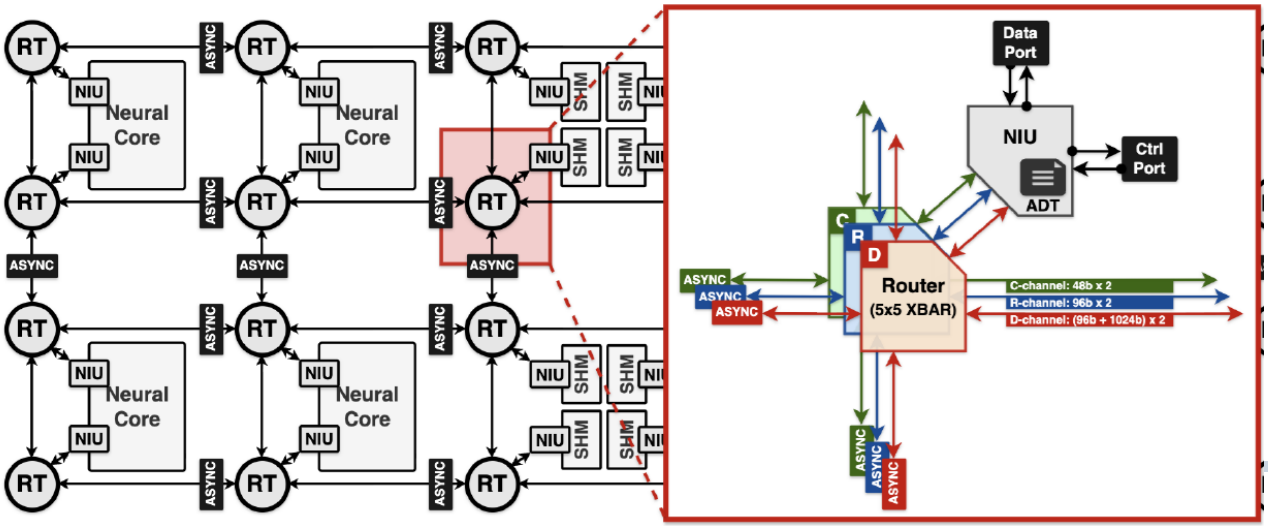
On-Chip Communication

- Evolution of interconnect structures
  - Shared bus (System bus)
  - Hierarchical bus
  - Bus matrix
  - Network on chip



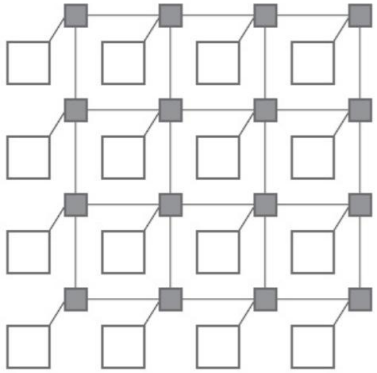
NoC Components

- Network on chip (NoC) is a packet switched on-chip communication network
- Data are carried in packets
- NoC consists of
  - Network interfaces (NI)
  - Switches (Routers)
  - Interconnects (Wires)



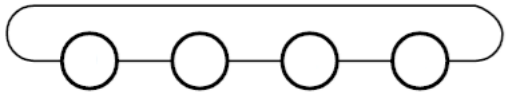
Topology

- Mesh
  - 2D mesh is the most popular

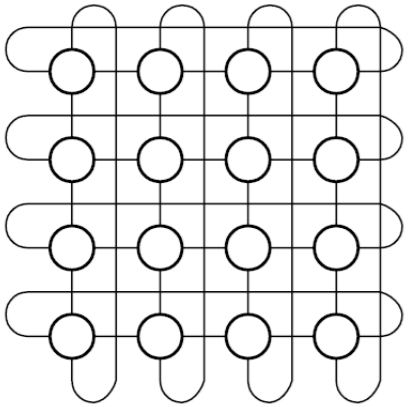


- Torus
  - 1D torus (ring)

    Example: CPU

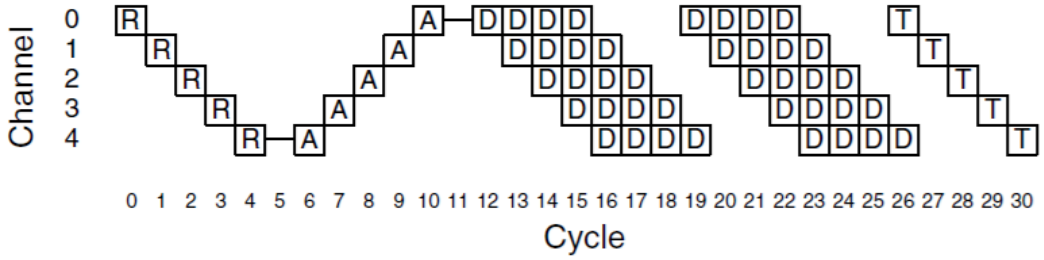


  - 2D torus

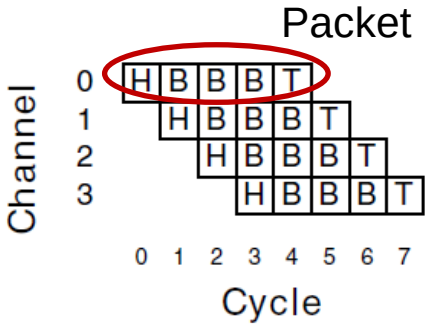


Switching

- Circuit switching
  1. Allocate a circuit from source to destination
  2. Send data along the circuit
  3. Deallocate the circuit
  - It forms a dedicated path



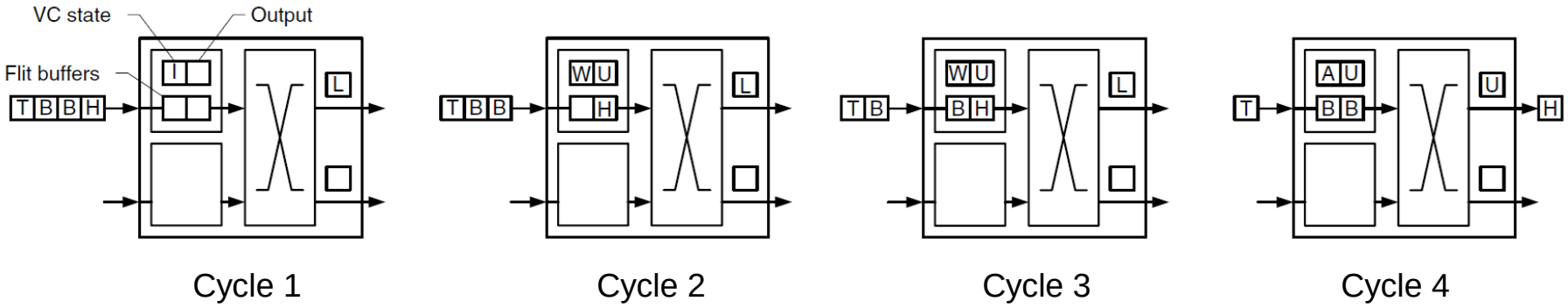
- Packet switching
  - Data is divided into packets
  - Each packet is routed independently
  - No dedicated path is required





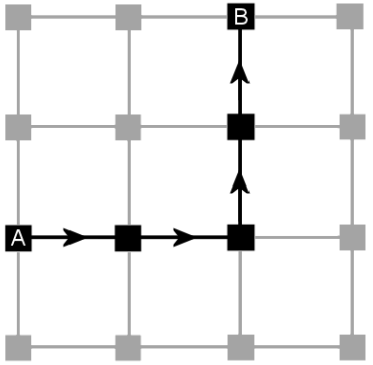
Packet Switching

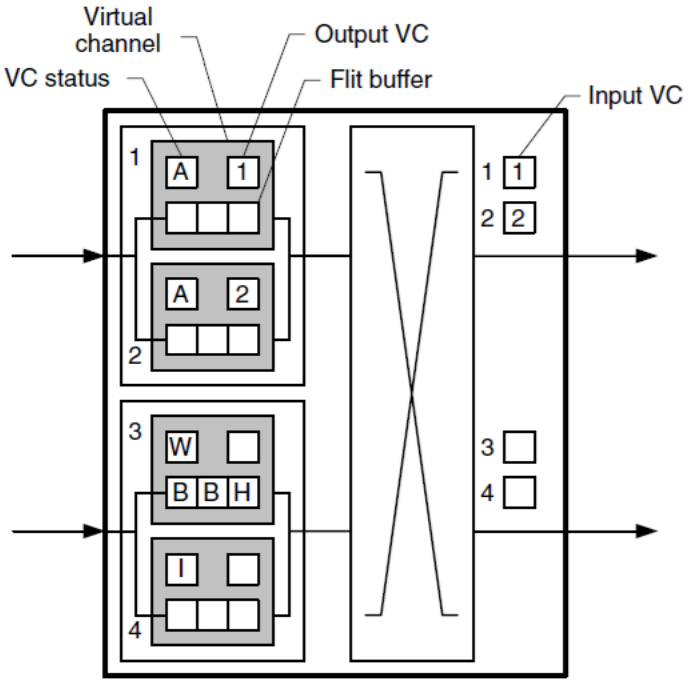
- A packet consists of a head flit, several body flits, and a tail flit
- When the head flit arrives at a node, it acquires
  - A virtual channel
  - One flit buffer and one flit of channel bandwidth
- When the body flit or tail flit arrives at a node, it acquires
  - One flit buffer and one flit of channel bandwidth



Routing

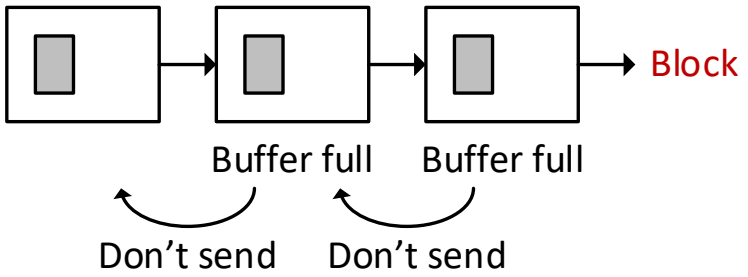
- Routing algorithms efficiently route packets from source to destination
- XY routing
  - It routes packets in the X-dimension first and then in the Y-dimension
- Router





Flow Control

- Upstream nodes are notified of the availability of buffers at the downstream nodes



- Credit-based flow control

